Aim: To install MongoDB and perform basic CRUD (Create, Read, Update, Delete) operations using MongoDB tools.

Q1. What is MongoDB?

A NoSQL database that stores data in flexible, JSON-like documents, Developed by MongoDB Inc. MongoDB is a cross-platform, open-source, database program that uses document-oriented data model. It stores data in flexible, JSON-like documents with dynamic schema, making integration easy and fast.

Q2. What is a Cluster?

A MongoDB cluster is a group of servers working together to manage and store database data. It provides high availability, scalability, and redundancy through data distribution.

Replication creates copies of data on different servers using replica sets for fault tolerance. If the primary server fails, a secondary server automatically takes over to maintain availability.

Q3. What is a Database?

A database in MongoDB is a container that holds groups of collections for organized data storage. Each database gets its own set of files on the file system.

Q4. What is a Collection?

A collection in MongoDB is a grouping of documents, analogous to a table in relational databases. It exists within a single database and does not enforce a strict schema.

Q5. What is a Document?

A document in MongoDB is a record in a collection, stored as a BSON (Binary JSON) object. It consists of field-and-value pairs, similar to JSON objects.

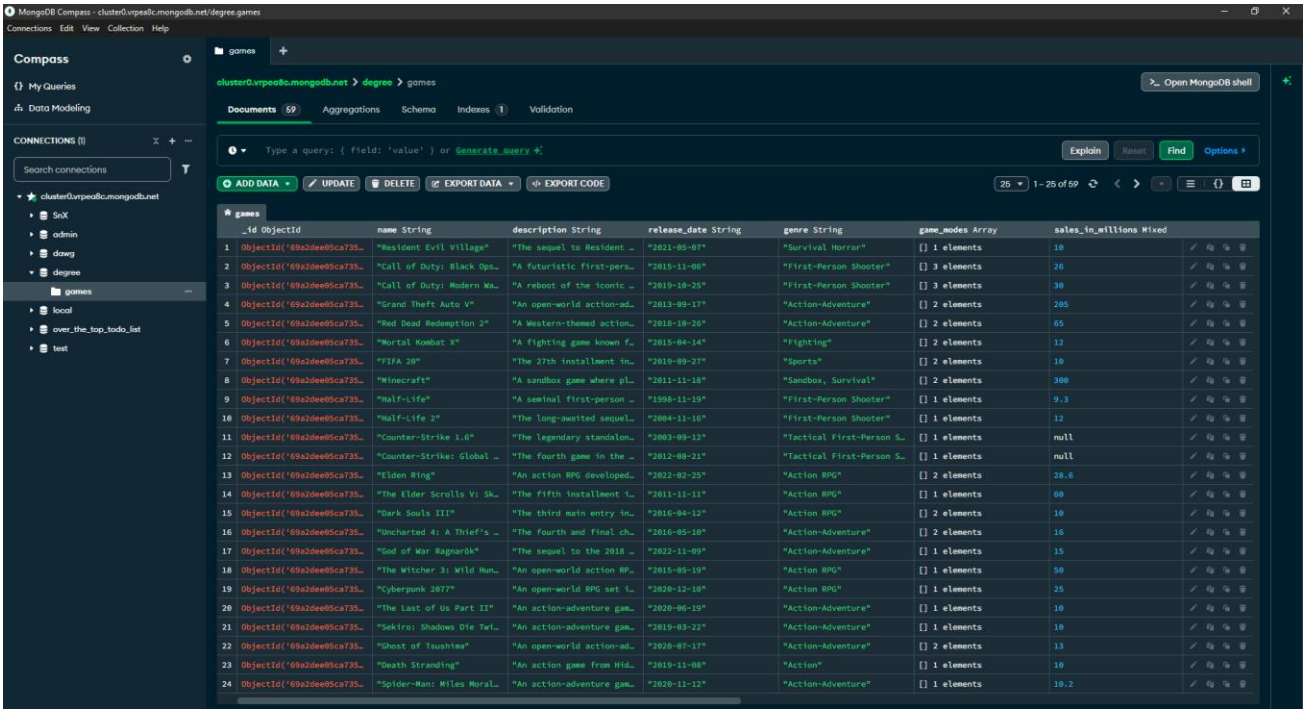
Q6. Steps to add/import data in MongoDB.

- Directly add documents to collections using `insertOne()` for single or `insertMany()` for multiple entries.
- Import JSON, CSV, or TSV files from the terminal using the `mongoimport` utility program.
- Use the intuitive graphical interface to manually add documents or drag-and-drop files for import.
- Copy data from existing collections using aggregation pipeline with `$out` or `$merge` stages.
- Insert data programmatically through official MongoDB drivers for languages like Python, Node.js, or Java.

Q6. List methods of connecting to MongoDB.

- Connect by passing a standard `URI` format to your `MongoDB` client or driver.
- Launch the command-line shell (`mongosh`) and specify the host and database directly.
- Establish a connection in code using official drivers like `PyMongo` or `Node.js`.
- Via MongoDB Compass GUI via a `URI`

Games data imported



`findOne()`: Returns a single document that matches the specified query criteria from a collection.

```
> db.games.findOne({ name: "Resident Evil 7: Biohazard" });
< {
    _id: ObjectId('69a2dee05ca73550295621cb'),
    name: 'Resident Evil 7: Biohazard',
    description: 'The seventh major installment in the Resident Evil series, returning to the survival horror roots with a first-person perspective. Players control Ethan Winters as he searches for his mi
    release_date: '2017-01-24',
    genre: 'Survival Horror',
    game_modes: [
      'Single-player'
    ],
    sales_in_millions: 13,
    developer: 'Capcom',
    publisher: 'Capcom',
    platforms: [
      'PC',
      'PlayStation 4',
      'Xbox One',
      'Nintendo Switch',
      'PlayStation 5',
      'Xbox Series X/S'
    ],
    age_rating: 'M (Mature 17+)',
    cover_art: 'https://example.com/covers/re7_biohazard.jpg'
  }
```

`deleteOne()`: Removes the first document that matches the specified filter from a collection.

```
> db.games.deleteOne({ name: "Resident Evil 7: Biohazard" });
< {
    acknowledged: true,
    deletedCount: 1
  }
```

`insertOne()`: Adds a single new document into a collection and returns an acknowledgement.

```
> db.games.insertOne({
    name: "Resident Evil 7: Biohazard",
    description: "The seventh major installment in the Resident Evil series, returning to the survival horror roots with a first-person perspective. Players control Ethan Winters as he searches fo
    release_date: "2017-01-24",
    genre: "Survival Horror",
    game_modes: ["Single-player"],
    sales_in_millions: 13.0,
    developer: "Capcom",
    publisher: "Capcom",
    platforms: ["PC", "PlayStation 4", "Xbox One", "Nintendo Switch", "PlayStation 5", "Xbox Series X/S"],
    age_rating: "M (Mature 17+)",
    cover_art: "https://example.com/covers/re7_biohazard.jpg"
});
< {
  acknowledged: true,
  insertedId: ObjectId('69a32d99aee5c4e9dc3468cc')
}
```

`deleteMany()`: Removes all documents that match the specified filter criteria from a collection.

```
> db.games.deleteMany({
      name: { $in: ["Horizon Zero Dawn", "Horizon Forbidden West"] }
  });
< {
    acknowledged: true,
    deletedCount: 2
  }
```

`insertMany()`: Inserts multiple documents into a collection in a single operation.

```
> db.games.insertMany([
    {
        name: "Horizon Zero Dawn",
        description: "An action role-playing game set in a post-apocalyptic world where tribal humans coexist with robotic creatures. Players control Aloy, a young outcast, as she embarks on a jou
        release_date: "2017-02-28",
        genre: "Action RPG",
        game_modes: ["Single-player"],
        sales_in_millions: 24.3,
        developer: "Guerrilla Games",
        publisher: "Sony Interactive Entertainment",
        platforms: ["PlayStation 4", "PC", "Nintendo Switch"],
        age_rating: "T (Teen)",
        cover_art: "https://example.com/covers/horizon_zero_dawn.jpg"
    },
    {
        name: "Horizon Forbidden West",
        description: "The sequel to Horizon Zero Dawn. Aloy travels to a forbidding stretch of the American coastline - the Forbidden West - to find the source of a mysterious plague that threaten
        release_date: "2022-02-18",
        genre: "Action RPG",
        game_modes: ["Single-player"],
        sales_in_millions: 8.4,
        developer: "Guerrilla Games",
        publisher: "Sony Interactive Entertainment",
        platforms: ["PlayStation 4", "PlayStation 5", "PC"],
        age_rating: "T (Teen)",
        cover_art: "https://example.com/covers/horizon_forbidden_west.jpg"
    }
]);
< {
  acknowledged: true,
  insertedIds: {
    '0': ObjectId('69a32f2daee5c4e9dc3468ce'),
    '1': ObjectId('69a32f2daee5c4e9dc3468cf')
  }
}
```

`updateOne()`: Modifies a single document that matches the filter criteria in a collection.

```
> db.games.findOne({ name: "Grand Theft Auto V" }, { name: 1, sales_in_millions: 1 });
< {
    _id: ObjectId('69a2dee05ca73550295621d1'),
    name: 'Grand Theft Auto V',
    sales_in_millions: 205
  }
> db.games.updateOne(
      { name: "Grand Theft Auto V" },
      { $set: { sales_in_millions: 215.0 } }
  );
< {
    acknowledged: true,
    insertedId: null,
    matchedCount: 1,
    modifiedCount: 1,
    upsertedCount: 0
  }
> db.games.findOne({ name: "Grand Theft Auto V" }, { name: 1, sales_in_millions: 1 });
< {
    _id: ObjectId('69a2dee05ca73550295621d1'),
    name: 'Grand Theft Auto V',
    sales_in_millions: 215
  }
```

`updateMany()`: Updates all documents matching the specified filter criteria in a collection.

```
> db.games.updateMany(
      { age_rating: "M (Mature 17+)" },
      { $set: { notes: "Contains mature content" } }
  );
< {
    acknowledged: true,
    insertedId: null,
    matchedCount: 37,
    modifiedCount: 37,
    upsertedCount: 0
  }
> db.games.find({ age_rating: "M (Mature 17+)" }, { name: 1, notes: 1 }).limit(3);
< {
    _id: ObjectId('69a2dee05ca73550295621cc'),
    name: 'Resident Evil Village',
    notes: 'Contains mature content'
  }
  {
    _id: ObjectId('69a2dee05ca73550295621cf'),
    name: 'Call of Duty: Black Ops III',
    notes: 'Contains mature content'
  }
  {
    _id: ObjectId('69a2dee05ca73550295621d0'),
    name: 'Call of Duty: Modern Warfare',
    notes: 'Contains mature content'
  }
```

Conclusion:

Successfully installed MongoDB and demonstrated fundamental database operations through practical CRUD command execution.